

No Genuine Degree-Three De Morgan Cohomology of the Dual-Rail Carrier, in Either Characteristic

Won Chul Yang

Independent researcher wcy0969@gmail.com

Public edition v1.0 (2026)

Abstract

The four-valued Belnap carrier D4 — the dual-rail lattice with states UNK, FAL, TRU, CON and order-reversing De Morgan involution τ — supports a single genuine essential cohomological obstruction: a degree-one, 2-primary class living in $H^1(V_4; \mathbb{F}_2) = \mathbb{F}_2^2$, the *carry*. A persistent question is whether the carrier also supports a *genuine degree-three* De Morgan obstruction. The natural candidate is the τ -antisymmetric 3-cochain $c_R^W(x, y, z)$, a second lattice difference of a τ -anti weight W , which is raw-nonzero (sup-norm 2). We answer the question in the negative, **in both characteristics**, and identify the mechanism. In characteristic 0 we prove a vanishing theorem: $H^3 = 0$ in each of the natural functorial carrier cochain theories — semilattice/Hochschild, group cohomology of $\langle \tau \rangle$ and V_4 , the order complex, the deformation core, and their τ -equivariant refinements — and we upgrade the list to a structural statement: the carrier’s building blocks are separable algebras (the semilattice part integrally, hence in every characteristic; the group part exactly when the characteristic is not 2), so every theory assembled from them vanishes above degree 0 in characteristic 0. We then pin the exact scope of this universality with an explicit counter-family: a normalized multi-term distributive homology theory, natural and functorial on the carrier, has $\dim_{\mathbb{Q}} H_3(\text{D4}; \mathbb{Q}) = 4$ — so the *unrestricted* universality claim is false, and the honest statement is universality over the separable/dual-admissible class. In characteristic 2 — the one place the ambient cohomology is genuinely nonzero ($H^3(V_4; \mathbb{F}_2)$ is 4-dimensional) — we prove the collapse theorem: an exact four-corner lattice identity forces $c_R^W \bmod 2 = (r+s)(\rho(x) + \rho(y))$, where ρ is the rail parity and (r, s) are the weight coordinates. The cochain is z -independent, is not a V_4 3-cocycle (explicit witness), defines no class in $H^3(V_4; \mathbb{F}_2)$, has vanishing full-module integral Bockstein class, and dies in the essential (nondegenerate-arity) quotient. The mechanism is uniform in both characteristics: the apparent cubic is the image of the degree-one carry parity under an arity-padding operator S_{12} , not a degree shift. Hence the carrier’s degree-three De Morgan arc is closed: the obstruction is rigid at degree one. All finite identities are machine-verified in Lean 4 (core only) and independently replayed; the two characteristic-0 rank computations are shipped as replayable exact-arithmetic certificates.

1 Introduction

The four-valued De Morgan carrier D4 is the bounded distributive lattice on the states $\{\text{UNK}, \text{FAL}, \text{TRU}, \text{CON}\}$, with the order-reversing involution τ (De Morgan complement). It is the semantic substrate of Belnap’s four-valued logic [1], and, in this series, of a dual-rail calculus tracking how information is *resolved* (the face $R = \{\text{TRU}, \text{FAL}\}$) versus *unresolved* ($\{\text{UNK}, \text{CON}\}$). Two companion

releases of this series establish the carrier’s complexity landscape: the exact negation-cost law $\nu = \text{dec}$ on the resolved face [13], and a cohomological reading of the negation-cost invariant as a character-tower length [14].

The carrier’s one genuine essential cohomological obstruction sits in degree **one**: the *carry*, a 2-primary class in

$$\text{Ext}_{\mathbb{F}_2[V_4]}^1(\text{triv}, \text{triv}) = H^1(V_4; \mathbb{F}_2) = \mathbb{F}_2^2,$$

where V_4 is the Klein four-group of dual-rail symmetries (§3). A recurring question one degree up is whether the carrier supports a *genuine degree-three* De Morgan obstruction. The natural candidate is the **τ -anti De Morgan 3-cochain**

$$c_R^W(x, y, z) = W((x \vee y) \vee z) - W((x \vee y) \wedge z) - W((x \wedge y) \vee z) + W((x \wedge y) \wedge z),$$

a second lattice difference of a weight $W : D4 \rightarrow \mathbb{Z}$ with $W(\tau t) = -W(t)$. The candidate is **raw-nonzero**: its sup-norm on $D4$ is 2 (Proposition 2.5). Does it represent a nonzero degree-3 class — a genuine “cubic carrier obstruction” one level above the carry?

This paper proves it does not — in either characteristic — and identifies the precise mechanism. The cubic is, uniformly, a *degenerate arity-3 padding* of the degree-1 carry parity: it equals the image $S_{12}(\rho)$ of the rail parity $\rho = a + b$ under the padding operator $S_{12}(u)(x, y, z) = u(x) + u(y)$. “Degree-3-looking” = padded degree-1.

The argument has two halves and a bridge.

- **Characteristic 0 (the vacuum, §4).** H^3 vanishes in every natural functorial carrier theory we name, so $[c_R^W] = 0$ there. We strengthen the named-family check to a *structural separability* statement, and — crucially — we **pin its exact scope** with an explicit counter-family, so the universality claim is honest: it holds over the separable/dual-admissible class, and provably not unrestrictedly.
- **Characteristic 2 (where it could survive, §5).** This is the only place Maschke’s theorem fails for V_4 , so $H^3(V_4; \mathbb{F}_2)$ is genuinely nonzero — yet we prove c_R^W *still* collapses, by an exact four-corner lattice identity, to padded degree-1.

The two halves combine into the main theorem (§7): no genuine essential degree-three carrier De Morgan cohomology in any characteristic. This supplies the *degree-axis rigidity* input for the series’ program of consolidating the carrier’s obstruction theory into a single degree-one class; the separations along other axes (the associator’s bidegree, the quantum-error-correction comparison) are treated in forthcoming companion releases and are not claimed here (§8).

Verification discipline. Every finite identity on $D4$ is proved in full in the text, machine-verified in Lean 4 (core only, no **sorry**, no **native_decide**; §9), and independently replayed by standalone scripts. Exactly two facts in the paper are *certificate-level computational premises* — the two characteristic-0 rank computations (Fact 4.11 and the order-complex ranks in Theorem 4.5(O), the latter also proved conceptually) — and they are stated as such, with exact-rational replay scripts shipped alongside (§9).

2 The carrier, the involution, and the cubic candidate

2.1 Dual-rail coordinates

Definition 2.1. The carrier $D4$ is the set $L = (\mathbb{F}_2)^2$ in *dual-rail coordinates* $u = (u_+, u_-)$:

$$\text{UNK} = (0, 0), \quad \text{TRU} = (1, 0), \quad \text{FAL} = (0, 1), \quad \text{CON} = (1, 1),$$

ordered componentwise, with meet $\wedge$ and join $\vee$ computed railwise (Boolean AND/OR per coordinate). This is the four-element Boolean lattice; UNK is the bottom, CON the top. The **De Morgan involution** is

$$\tau(u) = u + (1, 1) \quad (\text{bitwise NOT}),$$

which is order-reversing with $\tau^2 = \text{id}$, and swaps $\text{UNK} \leftrightarrow \text{CON}$, $\text{TRU} \leftrightarrow \text{FAL}$. The **resolved face** is $R = \{\text{TRU}, \text{FAL}\}$; the **rail parity** is the $\mathbb{F}_2$ -linear functional $\rho(u) = u_+ + u_-$, which equals 1 exactly on R .

Note that over $\mathbb{F}_2$ the involution τ is a *translation* of the group $(L, +)$, not a group automorphism; this is why the τ -anti objects below are not automatically objects of ordinary group cohomology, and why the failure in Theorem 5.6 has to be checked rather than defined away.

Throughout, V_4 denotes the Klein four-group. Two distinct roles of V_4 occur and must not be conflated: (i) the *state group* $(L, +) \cong V_4$ (used in §5.4, where D4's states are added railwise), and (ii) the *symmetry group* $V_4 = \langle \sigma, P \rangle$ of dual-rail symmetries generated by the De Morgan involution $\sigma = \tau$ and the monotone rail swap $P(u_+, u_-) = (u_-, u_+)$ (used in §3). Both are elementary abelian of order 4, which is exactly why the same cohomology $H^*(V_4; \mathbb{F}_2)$ appears in both roles.

2.2 τ -anti weights

Definition 2.2. A τ -anti weight is a function $W : L \rightarrow \mathbb{Z}$ with $W(\tau t) = -W(t)$ for all t . Define

$$W_A(\text{UNK}) = 1, W_A(\text{CON}) = -1, W_A(\text{TRU}) = W_A(\text{FAL}) = 0; \quad W_B(\text{TRU}) = 1, W_B(\text{FAL}) = -1, W_B(\text{UNK}) =$$

Lemma 2.3 (the weight space). *The τ -anti weights form a free $\mathbb{Z}$ -module of rank 2 with basis $\{W_A, W_B\}$: every τ -anti weight is $W = rW_A + sW_B$ with $r = W(\text{UNK})$, $s = W(\text{TRU})$, uniquely.*

Proof. τ pairs the states as $\{\text{UNK}, \text{CON}\}$ and $\{\text{TRU}, \text{FAL}\}$. Anti-symmetry determines W on each pair by its value on one member: $W(\text{CON}) = -W(\text{UNK})$, $W(\text{FAL}) = -W(\text{TRU})$. Hence W is freely determined by $(W(\text{UNK}), W(\text{TRU})) = (r, s)$, and this W equals $rW_A + sW_B$ by inspection on all four states. $\square$

2.3 The cubic candidate

Definition 2.4. For a τ -anti weight W , the **De Morgan 3-cochain** is

$$c_R^W(x, y, z) = W(J \vee z) - W(J \wedge z) - W(M \vee z) + W(M \wedge z), \quad J = x \vee y, \quad M = x \wedge y.$$

This is the alternating four-corner sum of W over the square $\{J, M\} \times \{\vee z, \wedge z\}$ — a second lattice (Newton) difference of W , first in the pair (x, y) , then in z .

Proposition 2.5 (raw facts). *On D4:*

1. $\|c_R^{W_A}\|_{\text{sup}} = \|c_R^{W_B}\|_{\text{sup}} = 2$; the exact value distribution of $c_R^{W_A}$ over the 64 triples is $\{-2: 4, -1: 16, 0: 24, 1: 16, 2: 4\}$, and likewise for W_B .
2. c_R^W is τ -antisymmetric: $c_R^W(\tau x, \tau y, \tau z) = -c_R^W(x, y, z)$.

3. c_R^W vanishes on $\text{Fix}(\tau)^3$ (vacuously: $\text{Fix}(\tau) = \emptyset$ on D4) and sums to zero over L^3 .
4. The mod-2 support of $c_R^{W_A}$ has exactly 32 of the 64 triples.

Proof. (2): τ is order-reversing, so it swaps $\vee \leftrightarrow \wedge$; hence τ maps the four corners of the defining sum for (x, y, z) to the four corners for $(\tau x, \tau y, \tau z)$ with the $+$ -signed and $-$ -signed corners exchanged; combined with $W(\tau t) = -W(t)$, the total sign is -1 . (1), (3), (4) are finite evaluations over the 64 triples; they are corroborated by the shipped replay (§9) and none of them is load-bearing for the sequel — the collapse theorem (Theorem 5.4) re-derives everything that matters. $\square$

The candidate is thus **raw-nonzero**. The entire content of the paper is that *raw-nonzero* $\neq$ *essential*: in every natural cohomological home, its class is zero or it fails to be a class at all.

3 The degree-one carry

Proposition 3.1. *Let $V_4 = \langle \sigma, P \rangle$ be the symmetry Klein four-group acting on the carrier, and let $\mathbb{F}_2$ carry the trivial action. Then*

$$H^1(V_4; \mathbb{F}_2) = \text{Hom}(V_4, \mathbb{F}_2) = \mathbb{F}_2^2,$$

with basis the two characters χ_σ, χ_P dual to the generators.

Proof. For trivial action, a 1-cocycle is a map $f : G \rightarrow \mathbb{F}_2$ with $f(gh) = f(g) + f(h)$ — a homomorphism — and the coboundaries are zero, so $H^1(G; \mathbb{F}_2) = \text{Hom}(G, \mathbb{F}_2)$. For $G = V_4 = C_2 \times C_2$ a homomorphism to $\mathbb{F}_2$ is freely determined by its values on the two generators. $\square$

Definition 3.2. The **carry** is the class $\chi_\sigma + \chi_P \in H^1(V_4; \mathbb{F}_2)$: the character which is 1 exactly on the symmetries acting nontrivially on the resolved face R . Its state-level shadow is the rail parity $\rho = a + b$ of Definition 2.1 (the indicator of R).

Two published anchors of the series meet this object from the complexity side: on the resolved face the negation cost satisfies $\nu(f) = \text{dec}(f)$ exactly [13], and dec is the length of a minimal nested χ_σ -character tower [14]; the $\text{dec} = 0 \rightarrow 1$ boundary is the complexity shadow of the χ_σ -component of the carry. Nothing below depends on more than Proposition 3.1; the anchors are context.

Remark 3.3. $H^1(V_4; \mathbb{F}_2) = \mathbb{F}_2^2$ is nonzero in every characteristic-2 setting, while all the characteristic-0 homes of §4 vanish. The degree-3 question is therefore only interesting in characteristic 2 — which is exactly where §5 works, *against* a nonzero ambient H^3 .

4 Characteristic zero: the vacuum, and its exact scope

Fix a field k of characteristic 0 (the reader may take $k = \mathbb{Q}$).

4.1 The Möbius algebra

Lemma 4.1 (Möbius algebra; cf. Solomon [4], Greene [5]). *Let $(L, \wedge)$ be a finite meet-semilattice and k any commutative ring. The semilattice algebra $k[L, \wedge]$ (free k -module on $\{\delta_x\}_{x \in L}$, product $\delta_x \delta_y = \delta_{x \wedge y}$) is isomorphic, as a k -algebra, to the product ring k^L . The isomorphism is defined over $\mathbb{Z}$.*

Proof. Define $\varphi : k[L, \wedge] \rightarrow k^L$ on the basis by $\varphi(\delta_y) = (\mathbf{1}[x \leq y])_{x \in L}$, extended k -linearly. Then $\varphi(\delta_y)\varphi(\delta_{y'}) = (\mathbf{1}[x \leq y] \cdot \mathbf{1}[x \leq y'])_x = (\mathbf{1}[x \leq y \wedge y'])_x = \varphi(\delta_{y \wedge y'})$ (using that $x \leq y$ and $x \leq y'$ iff $x \leq y \wedge y'$), so φ is a ring homomorphism. Its matrix in the bases $\{\delta_y\}$ and the coordinate idempotents of k^L is the incidence zeta matrix $\zeta(x, y) = \mathbf{1}[x \leq y]$; ordering L by a linear extension of $\leq$, ζ is upper unitriangular, hence invertible over $\mathbb{Z}$ (with inverse the Möbius matrix μ [6]). So φ is a k -algebra isomorphism for every k , defined over $\mathbb{Z}$. $\square$

Dually the same holds for $(L, \vee)$. The lemma is characteristic-*independent*: the semilattice building block splits over $\mathbb{F}_2$ as well. The characteristic-sensitivity of the carrier lives entirely in the group part (Lemma 4.4(2)).

4.2 Separability and the transfer lemma

Definition 4.2. A k -algebra A is **separable** if A is projective as a module over its enveloping algebra $A^e = A \otimes_k A^{\text{op}}$; equivalently, if there is a *separability idempotent* $e = \sum a_i \otimes b_i \in A^e$ with $\sum a_i b_i = 1$ and $(a \otimes 1)e = (1 \otimes a)e$ for all $a \in A$ [7].

Lemma 4.3 (separability kills Hochschild cohomology). *If A is separable then $\text{HH}^n(A, M) = 0$ for every A -bimodule M and every $n > 0$; the same holds for every cohomology of A computed as $\text{Ext}_{A^e}^n(A, -)$.*

Proof. $\text{HH}^n(A, M) = \text{Ext}_{A^e}^n(A, M)$ [8, Ch. IX]. If A is A^e -projective, $\text{Ext}_{A^e}^{>0}(A, -)$ vanishes identically. $\square$

Lemma 4.4 (the carrier's building blocks). *1. $k[L, \wedge]$ and $k[L, \vee]$ are separable for every field k (indeed k^L is separable with $e = \sum_x \delta_x \otimes \delta_x$, and Lemma 4.1 transports it).*

*2. $k[G]$ for $G \in \{\langle \tau \rangle \cong C_2, V_4\}$ is separable **iff** $\text{char } k \nmid |G|$, i.e. **iff** $\text{char } k \neq 2$; the idempotent is $e = |G|^{-1} \sum_g g \otimes g^{-1}$ (Maschke).*

3. Separability is preserved by finite tensor products: $e_{A \otimes B} = e_A \otimes e_B$.

Proof. (1) For $A = k^L$: $\sum \delta_x \delta_x = 1$ and $(\delta_y \otimes 1)e = \delta_y \otimes \delta_y = (1 \otimes \delta_y)e$; transport along the $\mathbb{Z}$ -defined isomorphism. (2) If $|G|$ is invertible, e as displayed satisfies both conditions by direct computation. Conversely if $\text{char } k = 2$ then $k[C_2] \cong k[t]/(t^2)$ is local non-semisimple, hence not separable (a separable algebra over a field is semisimple [7]); V_4 contains C_2 as a retract, and separability passes to retracts. (3) Direct check. $\square$

4.3 Theorem NG: the named-family vanishing

Theorem 4.5 (NG). *Let k have characteristic 0. Then $H^3 = 0$ — and in particular the class of c_R^W is zero — in each of the following natural functorial carrier cochain theories on D4:*

(B) semilattice/Hochschild: $\text{HH}^n(k[L, \wedge], M) = 0$ for all $n > 0$ and all bimodules M ; likewise for $(L, \vee)$.

- (G) group cohomology: $H^n(\langle\tau\rangle; k) = H^n(V_4; k) = 0$ for all $n > 0$.
- (O) order complex: $\Delta(D4)$ is contractible; $\tilde{H}^n(\Delta(D4); k) = 0$ for all n . Moreover $\Delta(D4)$ has f -vector $(4, 5, 2, 0)$ — there are no 3-simplices at all.
- (Op) deformation core: $HH^2 = HH^3 = 0$ for the carrier's characteristic-0 core algebra (any composite of the §4.2 blocks), so the Gerstenhaber deformation complex [9, 10] is rigid.
- (τ) equivariant refinements: the $\langle\tau\rangle$ -equivariant refinement of each theory above also has $H^3 = 0$.

Proof. (B): Lemmas 4.4(1) + 4.3. (G): Lemma 4.4(2) + 4.3, noting $H^n(G; k) = \text{Ext}_{k[G]}^n(k, k)$ and semisimplicity of $k[G]$ in characteristic 0. (O): $\Delta(D4)$ is the chain complex of the poset $D4$. Since $D4$ has a bottom element $\hat{0} = \text{UNK}$ comparable to everything, every chain extends by $\hat{0}$; hence $\Delta(D4)$ is a cone with apex $\hat{0}$, and cones are contractible. The f -vector is a direct count: 4 vertices; 5 comparable pairs ($\text{UNK} < \text{TRU}$, $\text{UNK} < \text{FAL}$, $\text{UNK} < \text{CON}$, $\text{TRU} < \text{CON}$, $\text{FAL} < \text{CON}$); 2 two-chains ($\text{UNK} < \text{TRU} < \text{CON}$, $\text{UNK} < \text{FAL} < \text{CON}$); no 3-chains. In particular $C^3 = 0$, so $H^3 = 0$ holds even before contractibility. (Op): Lemma 4.4(3) then 4.3. (τ): $|\langle\tau\rangle| = 2$ is invertible in k , so the averaging idempotent $(1 + \tau^*)/2$ splits each cochain complex into $\pm$ -isotypic summands and the equivariant (Borel) theory collapses onto the invariant summand of the underlying theory, with no higher group-cohomology contribution ($H^{>0}(C_2; k) = 0$ by (G)); a summand of zero is zero. $\square$

Remark 4.6 (the (Op) caveat: a trivial rigidity). The vanishing at (Op) is characteristic-0 *semisimplicity* — a trivial rigidity. It is **not** evidence of “no obstruction”: the genuine carry lives exactly where Maschke fails (characteristic 2). We cite (Op) only for the characteristic-0 vacuum.

Remark 4.7 (scope of Theorem NG, verbatim). NG is a no-go for the *named* families (B), (G), (O), (Op) and their τ -equivariant refinements, proved on $L = D4$. It is not a no-go for artificial two-term complexes built to make c_R^W a coboundary by fiat, and the *exhaustivity* of the named list among “all natural functorial theories” is precisely the universality question addressed — and bounded — in §4.5.

4.4 The vanishing is structural

Corollary 4.8 (coefficient-free closure). *In characteristic 0, every cohomology theory of every finite composite (tensor products, retracts) of the carrier's building blocks $\{\text{semilattice algebras}, k[\langle\tau\rangle], k[V_4]\}$ that is computed as Ext over the enveloping algebra vanishes in all degrees > 0 , for all coefficient bimodules. The sole characteristic-sensitive building block is the group algebra of the symmetry group — the home of the carry.*

Proof. Lemma 4.4 gives separability of each block in characteristic 0 and closure under $\otimes$; retracts of separable algebras are separable [7]. Lemma 4.3 kills all positive-degree Ext. The characteristic-sensitivity statement contrasts Lemma 4.1 (integral splitting) with Lemma 4.4(2). $\square$

4.5 The exact scope: an explicit counter-family

It is tempting to promote Theorem 4.5 to: “ $H^3 = 0$ in every natural functorial characteristic-0 carrier cochain theory.” One natural route is a factorization claim: every such theory factors through the carrier’s Priestley/Birkhoff dual [11]. The dual side is favorable:

Proposition 4.9 (dual reduction). *The Priestley/Birkhoff dual of D4 is the two-point antichain $J(\text{D4}) = \{\text{TRU}, \text{FAL}\}$ of join-irreducibles, with τ acting as the swap. Any characteristic-0 theory that factors naturally through the dual groupoid $[\{\text{TRU}, \text{FAL}\}/C_2]$ has $H^{>0} = 0$.*

Proof. The join-irreducibles of D4 are TRU and FAL (CON = TRU $\vee$ FAL is not irreducible; UNK is the bottom), mutually incomparable, and τ swaps them. The action groupoid of C_2 acting freely on a 2-point set is equivalent to a point; its category algebra is $M_2(k)$, which is separable, and Lemma 4.3 applies. $\square$

But the factorization premise is false without hypotheses — and hence the unrestricted universality claim is false. The witness is a multi-term distributive homology in the sense of Przytycki [2, 3].

Construction 4.10 (normalized multi-term distributive homology of D4). Encode $\text{D4} = \{0, 1, 2, 3\}$ by bitmask (UNK = 0, TRU = 1, FAL = 2, CON = 3). Consider the four binary operations

$$p_1(a, b) = a, \quad \vee, \quad \wedge, \quad p_2(a, b) = b,$$

each right-distributive over each of the others: for all $\star, \circ$ in the list, $(a \star b) \circ c = (a \circ c) \star (b \circ c)$.

Proof: for $\star = p_1$ (resp. p_2) both sides equal $a \circ c$ (resp. $b \circ c$); for $\circ = p_1$ both sides equal $a \star b$; for $\circ = p_2$ the left side is c and the right side is $c \star c = c$ by idempotence of every operation in the list; for $\star, \circ \in \{\vee, \wedge\}$ the four cases are lattice associativity+commutativity+idempotence or distributivity.

The *normalized chain group* C_n is the free $\mathbb{Q}$ -module on tuples $(t_0, \dots, t_n) \in \text{D4}^{n+1}$ with $t_i \neq t_{i+1}$; for an operation $\star$ the i -th face ($i \geq 1$) is

$$d_i^\star(t_0, \dots, t_n) = (t_0 \star t_i, \dots, t_{i-1} \star t_i, t_{i+1}, \dots, t_n),$$

$d_0(t_0, \dots, t_n) = (t_1, \dots, t_n)$, with tuples that become degenerate set to 0. Fix the coefficient vector $(1, -1, -1, 0)$ on $(p_1, \vee, \wedge, p_2)$ and set

$$d = \sum_i (-1)^i (d_i^{p_1} - d_i^\vee - d_i^\wedge).$$

Then $d^2 = 0$: this is the multi-term distributive-homology boundary of Przytycki [2] for a family of mutually right-distributive operations; it is also re-verified exactly, matrix-by-matrix, in the shipped replay. Write $H_n^N(\text{D4}; \mathbb{Q})$ for its homology.

Fact 4.11 (certificate: the rank table). *Over exact rational arithmetic: $\dim C_3 = 108$, $\text{rank } d_3 = 25$, $\text{rank } d_4 = 79$, so*

$$\dim_{\mathbb{Q}} H_3^N(\text{D4}; \mathbb{Q}) = 108 - 25 - 79 = 4, \quad \text{with } \tau\text{-eigenspace split } (+, -) = (2, 2);$$

dually $\dim_{\mathbb{Q}} H_N^3 = 4$ with τ -anti part of dimension 2. (Certificate-level computational premise: exact-rational rank computation, $d^2 = 0$ asserted matrix-wise, replayable by the shipped script; §9.)

Theorem 4.12 (the counter-family refutes unrestricted universality). *The theory $H_N^\bullet$ is natural and functorial on the carrier, τ acts on it, and it is not an artificial two-term complex; yet $H_N^3(D4; \mathbb{Q}) = 4 \neq 0$. Consequently:*

1. $H_N^\bullet$ does not factor through the Priestley dual;
2. the unrestricted claim “every natural functorial characteristic-0 carrier theory has $H^3 = 0$ ” is **false**;
3. Theorem 4.5’s universality is exactly over the **separable/dual-admissible class**: the named families and all composites of separable building blocks (Corollary 4.8) lie in it; $H_N^\bullet$ does not.

Proof. Naturality: a lattice homomorphism f commutes with each of the four operations (with p_1, p_2 trivially), hence with each face map, hence with d ; degeneracy is preserved; so f induces a chain map. τ -equivariance: τ is a lattice anti-homomorphism with $\tau\vee = \wedge\tau$ railwise; in d the $\vee$ - and $\wedge$ -terms carry the same coefficient (-1) , so conjugation by τ permutes them and fixes d ; the p_1 -term is fixed. Non-vanishing: Fact 4.11. Then (1) is the contrapositive of Proposition 4.9, (2) is witnessed by $H_N^\bullet$, and (3) restates Corollary 4.8 plus (2). $\square$

Remark 4.13 (honest scope, certificate-relative). Theorem 4.5 plus Corollary 4.8 delimit a *sufficient* admissible class; Theorem 4.12 shows the class cannot be enlarged to “all natural functorial theories”. We flag, additionally, that the *maximal* admissible class is underdetermined by naturality and functoriality alone: whether certain degenerate decorations of an admissible theory (e.g. mapping-cone paddings of identity functors) are “admissible” is a convention, not a theorem, and our scope statements are relative to the stated class. A restricted universality theorem with a sharp admissibility hypothesis remains open (§8).

Remark 4.14. The counter-family exhibits a nonzero ambient H^3 of a *different* cochain theory on the same lattice. It is **not** a claim that c_R^W itself is essential — c_R^W ’s class remains zero/absent in every home considered (and §5 kills it in characteristic 2 directly).

5 Characteristic two: where it could survive — and the collapse

5.1 The ambient cohomology is genuinely nonzero

Proposition 5.1. $H^*(V_4; \mathbb{F}_2) = \mathbb{F}_2[a, b]$ with $\deg a = \deg b = 1$; in particular $H^3(V_4; \mathbb{F}_2) = \langle a^3, a^2b, ab^2, b^3 \rangle$ is 4-dimensional, and $H^1(V_4; \mathbb{F}_2) = \mathbb{F}_2^2$ houses the carry.

Proof. Standard: for an elementary abelian 2-group of rank n , $H^*((C_2)^n; \mathbb{F}_2)$ is polynomial on n degree-1 generators [12, Ch. II–III]; take $n = 2$. $\square$

Corollary 5.2. By §4, a genuine degree-3 carrier obstruction could only exist in characteristic 2. This is what makes the collapse below non-trivial: the cubic must be shown to die despite a nonzero ambient H^3 .

5.2 The four-corner identity

Lemma 5.3 (four-corner identity). In $(L, +) = (\mathbb{F}_2)^2$, for all $x, y, z \in L$, with $J = x \vee y$, $M = x \wedge y$:

$$(J \vee z) + (J \wedge z) + (M \vee z) + (M \wedge z) = x + y.$$

Proof. Railwise, for bits $p, r \in \mathbb{F}_2$: $p \vee r = p + r + pr$ and $p \wedge r = pr$, so $(p \vee r) + (p \wedge r) = p + r$. Applying this per rail to the pairs (J, z) and (M, z) :

$$(J \vee z) + (J \wedge z) + (M \vee z) + (M \wedge z) = (J + z) + (M + z) = J + M.$$

Again railwise, $J + M = (x \vee y) + (x \wedge y) = x + y$. $\square$

5.3 The collapse theorem

Theorem 5.4 (collapse). *Let $W = rW_A + sW_B$ be any τ -anti weight (Lemma 2.3). Then, identically on $D4^3$,*

$$c_R^W \bmod 2 = (r + s) (\rho(x) + \rho(y)).$$

*In particular $c_R^W \bmod 2$ is **independent of** z , and equals the arity-padding $S_{12}(\bar{\rho})$ of the rail parity, scaled by $(r+s) \bmod 2$, where $S_{12}(u)(x, y, z) := u(x) + u(y)$.*

Proof. Write $\bar{W} = W \bmod 2 : L \rightarrow \mathbb{F}_2$. From the definitions, $\bar{W}_A = 1 + \rho$ (it is 1 exactly on $\{\text{UNK}, \text{CON}\}$, the complement of R) and $\bar{W}_B = \rho$. Hence

$$\bar{W} = \bar{r}(1 + \rho) + \bar{s}\rho = \bar{r} + (\bar{r} + \bar{s})\rho,$$

an *affine* function of the $\mathbb{F}_2$ -linear functional ρ . Mod 2 the signs in Definition 2.4 disappear, so with $a_1 = J \vee z$, $a_2 = J \wedge z$, $a_3 = M \vee z$, $a_4 = M \wedge z$:

$$c_R^W \bmod 2 = \sum_{i=1}^4 \bar{W}(a_i) = 4\bar{r} + (\bar{r} + \bar{s}) \sum_{i=1}^4 \rho(a_i) = (\bar{r} + \bar{s}) \rho\left(\sum_{i=1}^4 a_i\right) = (\bar{r} + \bar{s}) \rho(x + y),$$

using linearity of ρ and the four-corner identity (Lemma 5.3); and $\rho(x + y) = \rho(x) + \rho(y)$. $\square$

Corollary 5.5 (degenerate arity). *$c_R^W \bmod 2$ lies in the image of $S_{12} : \text{Map}(L, \mathbb{F}_2) \rightarrow \text{Map}(L^3, \mathbb{F}_2)$; it depends on (x, y) only, and only through the degree-one datum ρ .*

5.4 It is not a cocycle, and defines no cubic class

Identify L with the state group $V_4 = (\mathbb{F}_2)^2$ under railwise addition, and let $\eta := c_R^W \bmod 2$ for any W with $r + s$ odd — i.e. $\eta(x, y, z) = \rho(x) + \rho(y)$ — regarded as an inhomogeneous 3-cochain on the group V_4 with trivial $\mathbb{F}_2$ -coefficients. The bar differential is

$$(\delta\eta)(g_1, g_2, g_3, g_4) = \eta(g_2, g_3, g_4) + \eta(g_1 + g_2, g_3, g_4) + \eta(g_1, g_2 + g_3, g_4) + \eta(g_1, g_2, g_3 + g_4) + \eta(g_1, g_2, g_3).$$

Theorem 5.6 (not a 3-cocycle). $\delta\eta \neq 0$; an explicit witness is

$$(\delta\eta)(\text{TRU}, \text{TRU}, \text{CON}, \text{TRU}) = 1.$$

Consequently η is not a 3-cocycle of the state group, and defines **no** class in $H^3(V_4; \mathbb{F}_2)$ — in particular it is none of the cubic classes a^3, a^2b, ab^2, b^3 .

Proof. With $\rho(\text{TRU}) = \rho(\text{FAL}) = 1$, $\rho(\text{UNK}) = \rho(\text{CON}) = 0$ and $\text{TRU} + \text{TRU} = \text{UNK}$, $\text{TRU} + \text{CON} = \text{CON} + \text{TRU} = \text{FAL}$:

$$\eta(\text{TRU}, \text{CON}, \text{TRU}) = 1, \quad \eta(\text{UNK}, \text{CON}, \text{TRU}) = 0, \quad \eta(\text{TRU}, \text{FAL}, \text{TRU}) = 0, \quad \eta(\text{TRU}, \text{TRU}, \text{FAL}) = 0, \quad \eta(\text{TRU}, \text{TRU}, \text{TRU}) = 0,$$

and the five-term sum is 1. (The full $\delta\eta$ table has 128 nonzero entries, 36 of them on identity-free 4-tuples; the shipped replay enumerates them.) $\square$

5.5 The integral Bockstein class vanishes

At the integral level $c := c_R^W$ satisfies $\tau c = -c$, so relative to $0 \rightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \rightarrow \mathbb{F}_2 \rightarrow 0$ there is a visible cochain-level Bockstein representative $(\tau c - c)/2 = -c$. The correct home for the resulting *class* is the τ -equivariant cohomology of the full lattice module, and there it dies:

Theorem 5.7 (no full-module Bockstein class). *The diagonal τ -action on L^3 is free, so $\mathbb{Z}[L^3] \cong \mathbb{Z}[C_2]^{32}$ as a C_2 -module. Hence $H^p(C_2; \mathbb{Z}[L^3]) = 0$ for all $p > 0$, and the Bockstein image of $[\eta]$ in the full-module theory is 0.*

Proof. $\tau u = u + (1, 1) \neq u$ for every u , so no point of L^3 is fixed and the 64 points fall into 32 free orbits; choosing one point per orbit exhibits $\mathbb{Z}[L^3]$ as free of rank 32 over $\mathbb{Z}[C_2]$. Cohomology with coefficients in a free (hence induced) module vanishes in positive degrees (Shapiro’s lemma [8, Ch. XII]). Any Bockstein class of the full-module theory lives in such a group, hence is 0. $\square$

5.6 The essential quotient kills it

Definition 5.8. Let $C_{\text{set}}^3 = \text{Map}(L^3, \mathbb{F}_2)$ and let the **degenerate submodule** $D^3 \subset C_{\text{set}}^3$ be the span of all cochains that factor through a proper coordinate projection $L^3 \rightarrow L^S$, $S \subsetneq \{1, 2, 3\}$. The **essential quotient** is $C_{\text{ess}}^3 = C_{\text{set}}^3 / D^3$: it retains exactly the genuinely 3-variable content of a cochain.

Corollary 5.9. $\eta = c_R^W \bmod 2$ factors through (x, y) (Theorem 5.4), so $\eta \in D^3$ and $[\eta] = 0$ in C_{ess}^3 . The candidate has no genuinely ternary content at all.

6 The mechanism: arity padding, not a degree shift

The operator realizing the cubic from the degree-one datum is

$$S_{12} : \text{Map}(L, \mathbb{F}_2) \rightarrow \text{Map}(L^3, \mathbb{F}_2), \quad S_{12}(u)(x, y, z) = u(x) + u(y),$$

and Theorem 5.4 says $c_R^W \bmod 2 = (r+s) S_{12}(\rho)$. Three standard degree-raising constructions are thereby *excluded* as explanations of the cubic, each for a proven reason:

1. **Not a cup product.** A cup product of degree-1 classes of $H^1(V_4; \mathbb{F}_2)$ is a 3-cocycle representing an element of $\langle a^3, a^2b, ab^2, b^3 \rangle$; η is not a cocycle at all (Theorem 5.6).
2. **Not Tate/periodicity.** Any periodicity image of a class is a class; η defines no class.
3. **Not a Bockstein.** The full-module integral Bockstein class vanishes (Theorem 5.7).

What remains is exactly what S_{12} is: a *degenerate arity-3 padding* of degree-one data (Corollary 5.9). “Degree-3-looking” = padded degree-1.

7 Main theorem

Theorem 7.1 (no genuine degree-three De Morgan cohomology). *On the dual-rail carrier D4, there is no genuine essential degree-3 De Morgan obstruction, in any characteristic. Precisely: for every τ -anti weight W , the cubic candidate c_R^W*

1. *has zero class in every characteristic-0 theory of the separable/dual-admissible class — the named families (B) , (G) , (O) , (Op) and their τ -equivariant refinements included (Theorem 4.5, Corollary 4.8) — with the admissible-class boundary exhibited sharply by the distributive-homology counter-family (Theorem 4.12);*
2. *in characteristic 2, where the ambient $H^3(V_4; \mathbb{F}_2) \neq 0$, collapses identically to $(r+s) S_{12}(\rho)$: z -independent, not a 3-cocycle (explicit witness), defining no class in $H^3(V_4; \mathbb{F}_2)$, with vanishing full-module Bockstein class, and vanishing in the essential quotient (Theorems 5.4–5.7, Corollary 5.9).*

*In both characteristics the realizing mechanism is the arity-padding S_{12} of the degree-one carry parity ρ — not a cup product, not periodicity, not a Bockstein (§6). The carrier’s degree-3 De Morgan arc is closed: **the obstruction is rigid at degree one.***

Proof. Part 1 is §4; part 2 is §5; the mechanism statements are §6, each reduced to a proved theorem. $\square$

8 Scope, and what is not claimed

- **D4 only.** Every computation and theorem above is on the four-element carrier (or its two symmetry/state groups V_4). Nothing is claimed for larger De Morgan lattices; whether degree-one rigidity persists at B_n , $n \geq 3$, is **open**, and a higher essential class there would be *consistent* with Theorem 7.1.
- **Only this cubic.** The theorem closes the carrier’s δ^2 -type De Morgan candidate c_R^W . Other cubic functionals on other complexes are different objects and are not addressed.
- **The admissible-class boundary is part of the theorem.** The characteristic-0 universality is exactly over the separable/dual-admissible class; the unrestricted universality is *refuted*, not merely unproved; and the maximal admissible class is underdetermined by naturality alone (Remark 4.13). A restricted universality theorem with a sharp admissibility hypothesis is an open problem.
- **Degree versus level.** The negation-cost filtration $\text{dec}(f)$ of the companion releases [13, 14] is a *level* (complexity) parameter, not a cohomological degree; nothing here shifts the carry out of degree 1. The separation of the carry from the associator (a bidegree statement) and its comparison with quantum-error-correction Bockstein towers are the subjects of forthcoming companion releases and are **not claimed here**.
- **Size $\neq$ obstruction.** The exact clone counts of the series (e.g. the ternary selector count 103,275 on the odd flat carrier [15]) are cardinalities of a different V_4 -governed object and play no role here.

9 Mechanization and verification

Lean (structural spine). All finite identities on D4 are machine-verified in the Lean 4 library `Demorgan3` (core only; no `mathlib`, no `native_decide`, no `sorry`, no `axiom` declarations; 24 theorems, 22 axiom-free, 2 with kernel axiom profile exactly `[propext, Quot.sound]`; no classical choice anywhere): the weight-space lemma; the classification $\text{Hom}(V_4, \mathbb{F}_2) = \mathbb{F}_2^2$; the

four-corner identity (64 triples); the Boolean collapse for all parity pairs, its agreement with the integer mod-2 reduction, and z -independence; the integer-level collapse $2 \mid (c_R^W(x, y, z) - (W(\text{UNK}) + W(\text{TRU}))(\rho x + \rho y))$ for an arbitrary τ -anti weight; the non-cocycle witness and $\delta\eta \neq 0$; freeness of the diagonal τ -action on L^3 ; and the order-complex counts (f -vector $(4, 5, 2, 0)$, no 3-chains, $\chi = 1$, cone premise). See `lean/demorgan3/` and the frozen audit `axcheck.log`.

Replay (independent). Two standalone exact-arithmetic scripts (Python, stdlib only) re-verify, with frozen outputs shipped alongside: `replay_char2_collapse.py` (raw facts, four-corner, collapse for seven integer coordinate pairs, z -independence, the full $\delta\eta$ table, the 32 free τ -orbits; ALL PASS) and `replay_char0_certificates.py` (the order-complex $\mathbb{Q}$ -cohomology and the distributive-homology certificate: mutual distributivity, $d^2 = 0$ matrix-wise, rank table $108/25/79$, $\dim H_3 = 4$, τ -split $(2, 2)$, computed over `fractions.Fraction` independently of the programme-internal computation it reproduces; ALL PASS).

The characteristic-0 rank computation is a *certificate-level premise* of Fact 4.11 only; every other statement is proved in the text, and the finite identities are additionally kernel-checked.

Authorship and provenance

The mathematics is the author's, developed inside the author's carrier programme. AI assistance (Anthropic Claude family) was used for the machine-verification layer — the Lean 4 development, the replay scripts, and manuscript preparation — with the Lean 4 kernel as the acceptance gate for that layer. This is distinct from the two earlier AI-collaborative releases of the series, which are labelled as such. Corrections are invited.

References

- [1] N. D. Belnap, *A useful four-valued logic*, in: J. M. Dunn, G. Epstein (eds.), *Modern Uses of Multiple-Valued Logic*, Reidel, 1977, pp. 5–37. doi:10.1007/978-94-010-1161-7_2
- [2] J. H. Przytycki, *Distributivity versus associativity in the homology theory of algebraic structures*, *Demonstratio Math.* 44 (2011), no. 4, 823–869. doi:10.1515/dema-2013-0337, arXiv:1109.4850
- [3] J. H. Przytycki, K. K. Putyra, *Homology of distributive lattices*, *J. Homotopy Relat. Struct.* 8 (2013), 35–65. doi:10.1007/s40062-012-0012-5
- [4] L. Solomon, *The Burnside algebra of a finite group*, *J. Combin. Theory* 2 (1967), 603–615. doi:10.1016/S0021-9800(67)80064-4
- [5] C. Greene, *On the Möbius algebra of a partially ordered set*, *Adv. Math.* 10 (1973), 177–187. doi:10.1016/0001-8708(73)90106-0
- [6] G.-C. Rota, *On the foundations of combinatorial theory I: Theory of Möbius functions*, *Z. Wahrscheinlichkeitstheorie* 2 (1964), 340–368. doi:10.1007/BF00531932
- [7] F. DeMeyer, E. Ingraham, *Separable Algebras over Commutative Rings*, *Lecture Notes in Math.* 181, Springer, 1971. doi:10.1007/BFb0061226
- [8] H. Cartan, S. Eilenberg, *Homological Algebra*, Princeton Univ. Press, 1956.

- [9] M. Gerstenhaber, *The cohomology structure of an associative ring*, Ann. of Math. 78 (1963), 267–288. doi:10.2307/1970343
- [10] M. Gerstenhaber, *On the deformation of rings and algebras*, Ann. of Math. 79 (1964), 59–103. doi:10.2307/1970484
- [11] H. A. Priestley, *Representation of distributive lattices by means of ordered Stone spaces*, Bull. London Math. Soc. 2 (1970), 186–190. doi:10.1112/blms/2.2.186
- [12] A. Adem, R. J. Milgram, *Cohomology of Finite Groups*, 2nd ed., Grundlehren Math. Wiss. 309, Springer, 2004. doi:10.1007/978-3-662-06280-7
- [13] W. C. Yang, *The Price of NOT on D_4* , 2026. <https://github.com/ycmath/price-of-not-on-d4>, doi:10.5281/zenodo.21800033
- [14] W. C. Yang, *The Cohomological Price of NOT*, 2026. <https://github.com/ycmath/cohomological-price-of-not>, doi:10.5281/zenodo.21775055
- [15] W. C. Yang, *Exact Selector Counting on the Odd Flat Carrier*, 2026. <https://github.com/ycmath/odd-flat-selector-count>, doi:10.5281/zenodo.21868044

Series hub: <https://github.com/ycmath/dual-rail-carrier-program>